INCH-POUND

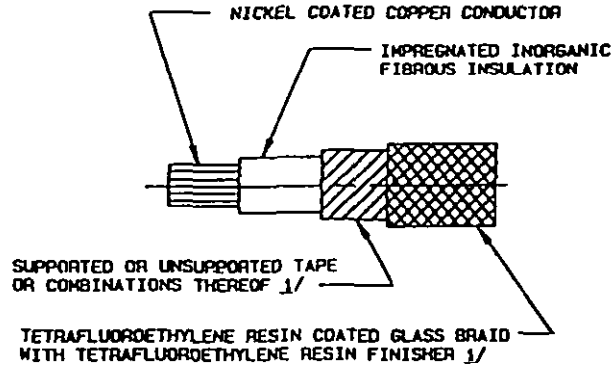
MIL-W-25038/1C
20 September 1991
SUPERSEDING
MIL-W-25038/1B
12 DEC 1980

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, HIGH TEMPERATURE AND FIRE RESISTANT, NORMAL WEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-25038.



1/ The outer two layers may be reversed in their sequence. The reversal is optional.

FIGURE 1. Wire, electrical.

REQUIREMENTS:

Design and construction: See figure 1.

Temperature rating: 260°C.

Voltage rating: 600 V rms.

Color: Unless otherwise specified by the Part or Identifying Number (PIN), the color shall be white.

Thermal shock: Maximum change in insulation measurement:

© 0.060 inch for sizes 22 through 10;
0.125 inch for sizes 8 through 4;
0.187 inch for sizes 0 through 0000.

© Denotes change

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(C) Durability of identification: 250 cycles (500 strokes) with 150 gram weight.

(C) PIN: M25038/1- (wire size from table I)-(color code designator in accordance with MIL-STD-681).

Examples:
Size 20, white - M25038/1-20-9;
Size 20, white with orange stripe - M25038/1-20-93.

(C) NOTE: Materials used in construction of the wire must be identified and approved by the qualifying activity. Wire construction cannot contain asbestos.

TABLE I. Finished wire construction and performance details.

Finished wire construction				Performance details	
				Bend test and cold bend	
Wire size	Resistance at 20°C (68°F) ohms/1000 ft (max)	Overall diameter in inches	Weight lb/1000 ft (max)	Mandrel diameter in inches	Test load (lb)
22 1/	23.7	.108 ± .008	10.0	6	2
20 1/	14.6	.117 ± .008	12.0	6	2
18 1/	9.14	.127 ± .008	15.0	6	2
16 1/	6.85	.137 ± .010	19.0	8	3
14 1/	4.32	.160 ± .010	25.0	8	3
12 1/	2.78	.175 ± .010	35.0	8	3
10	1.68	.220 ± .010	55.0	10	5
8	.936	.268 ± .012	85.0	12	5
6	.591	.330 ± .012	127	14	10
4	.375	.395 ± .012	192	16	10
2	.241	.472 ± .012	291	20	15
1	.196	.515 ± .018	347	22	15
0	.153	.555 ± .018	415	24	20
00	.120	.615 ± .020	520	24	25
000	.096	.680 ± .20	648	24	30
0000	.077	.750 ± .20	793	24	30

(C) 1/ Wire size 22 through 12 are inactive for new design. For new design use MIL-W-25038/3.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-1168)

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